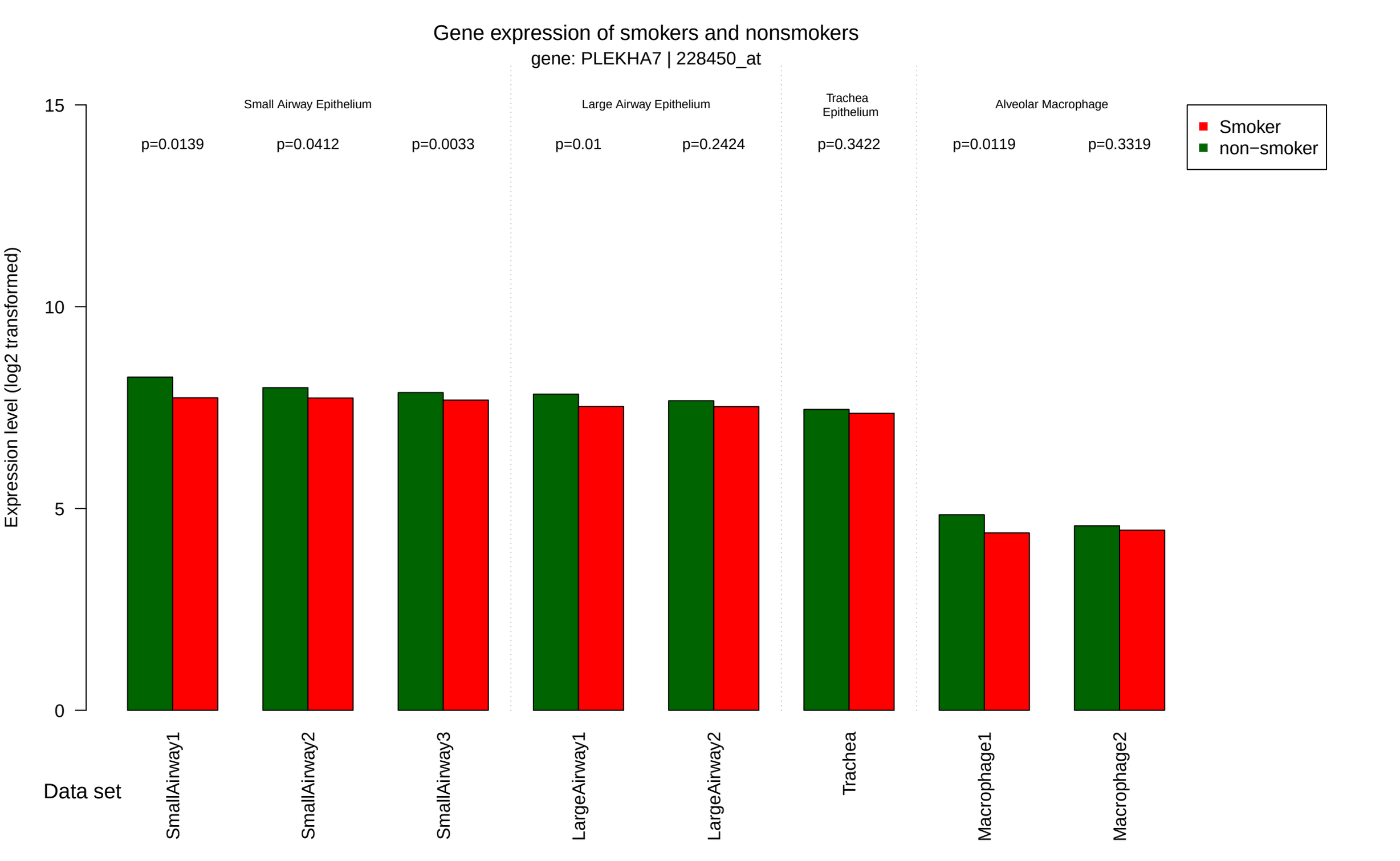
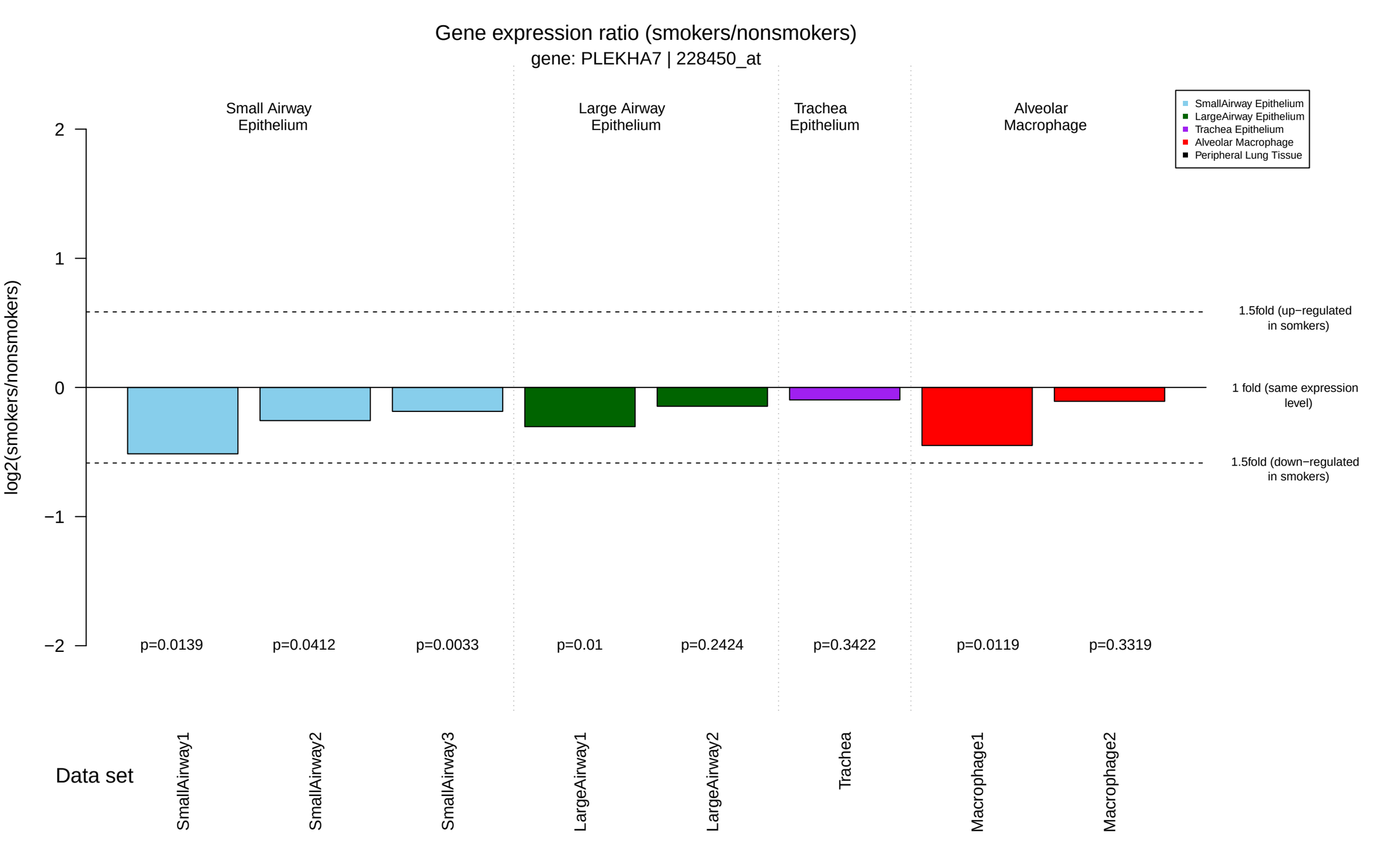
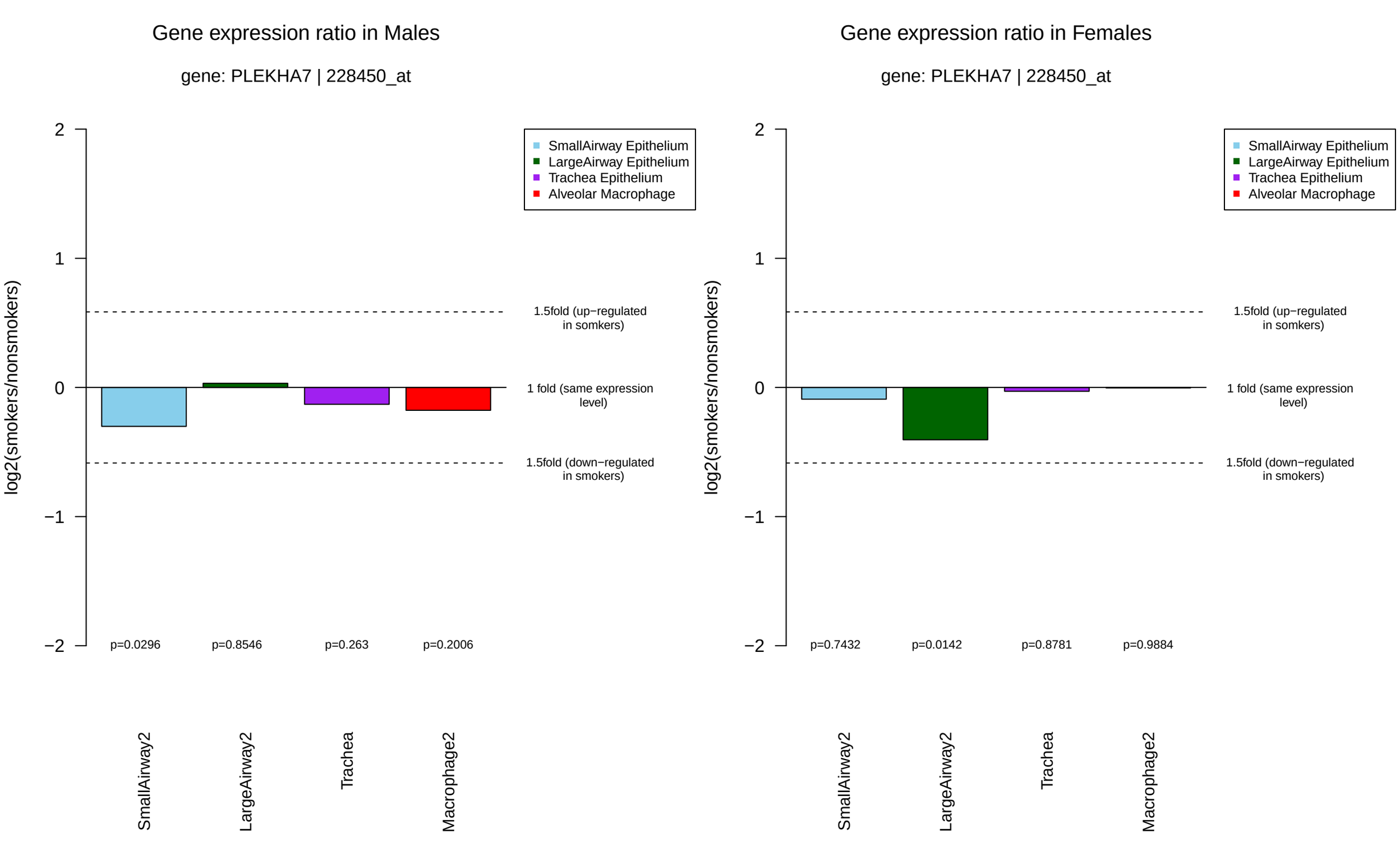
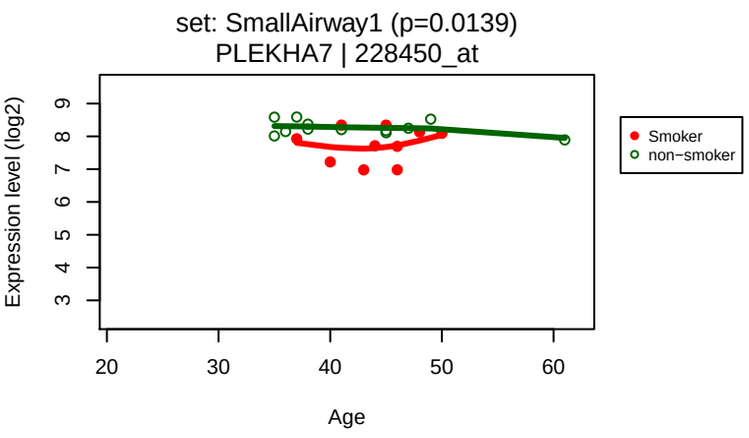
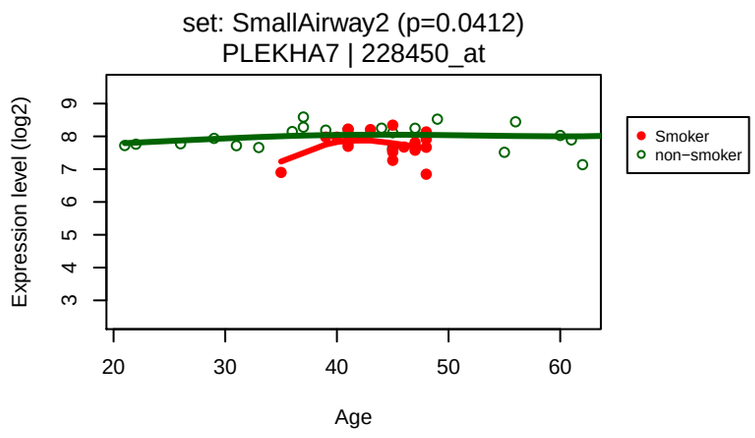
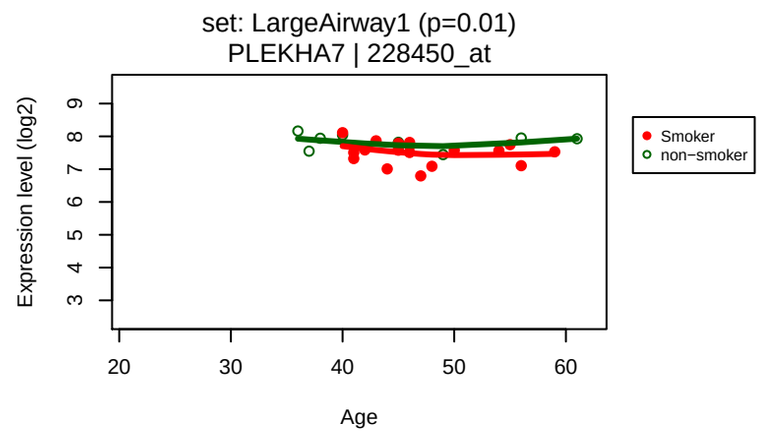

Gene expression of smokers and nonsmokers
gene: PLEKHA7 | 228450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0139
p=0.0412
p=0.0033
p=0.01
p=0.2424
p=0.3422
p=0.0119
p=0.3319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

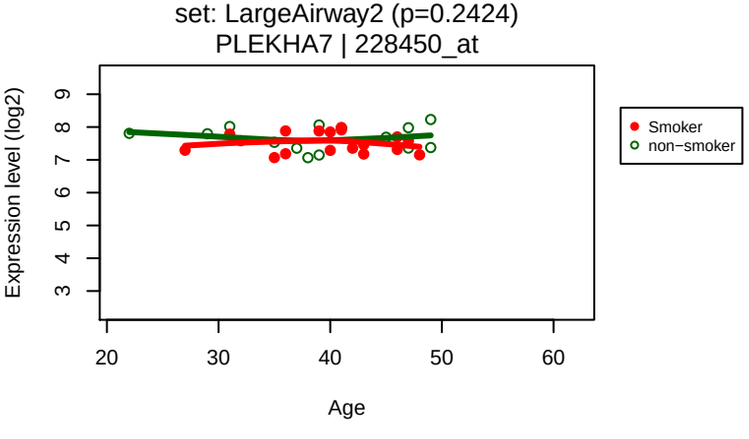

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA7 | 228450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0139
p=0.0412
p=0.0033
p=0.01
p=0.2424
p=0.3422
p=0.0119
p=0.3319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

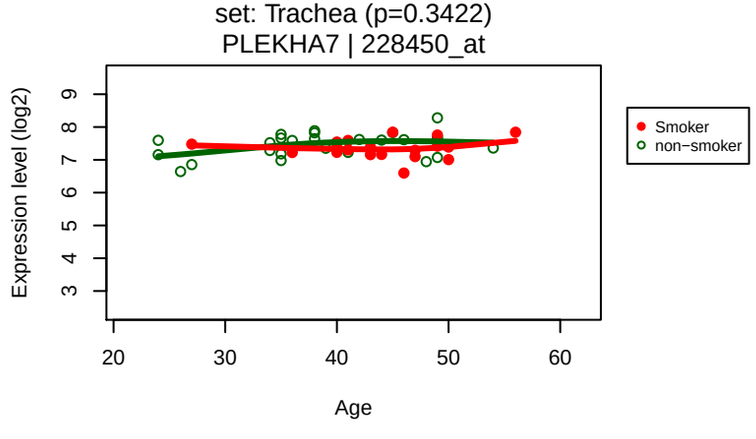

Gene expression ratio in Males
gene: PLEKHA7 | 228450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0296
p=0.8546
p=0.263
p=0.2006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA7 | 228450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7432
p=0.0142
p=0.8781
p=0.9884
SmallAirway2
LargeAirway2
Trachea
Macrophage2

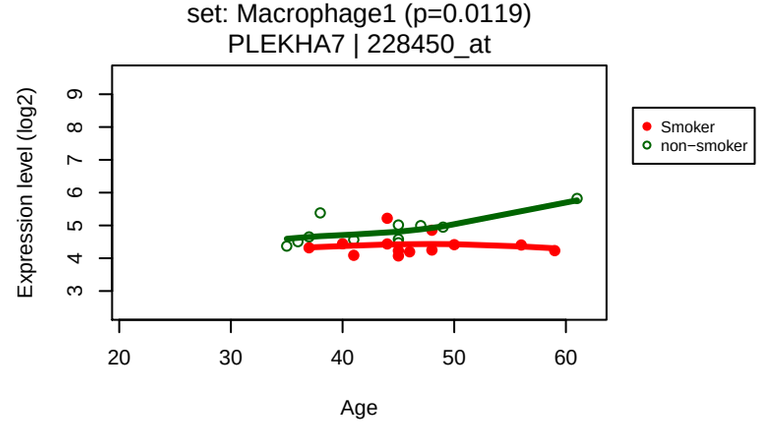

set: SmallAirway1 (p=0.0139)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker

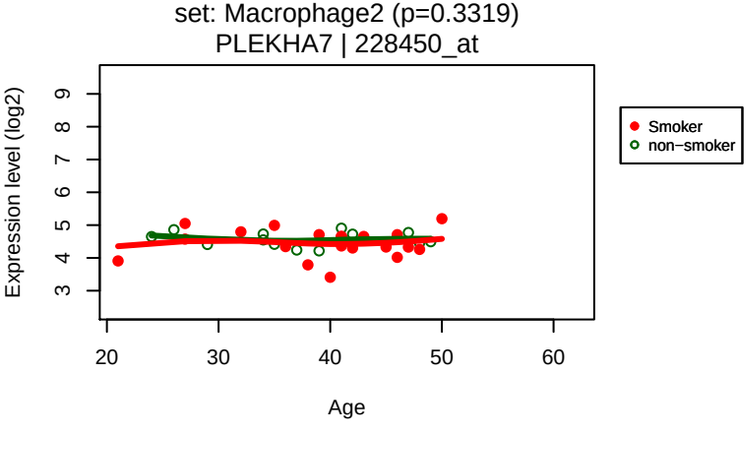

set: SmallAirway2 (p=0.0412)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker

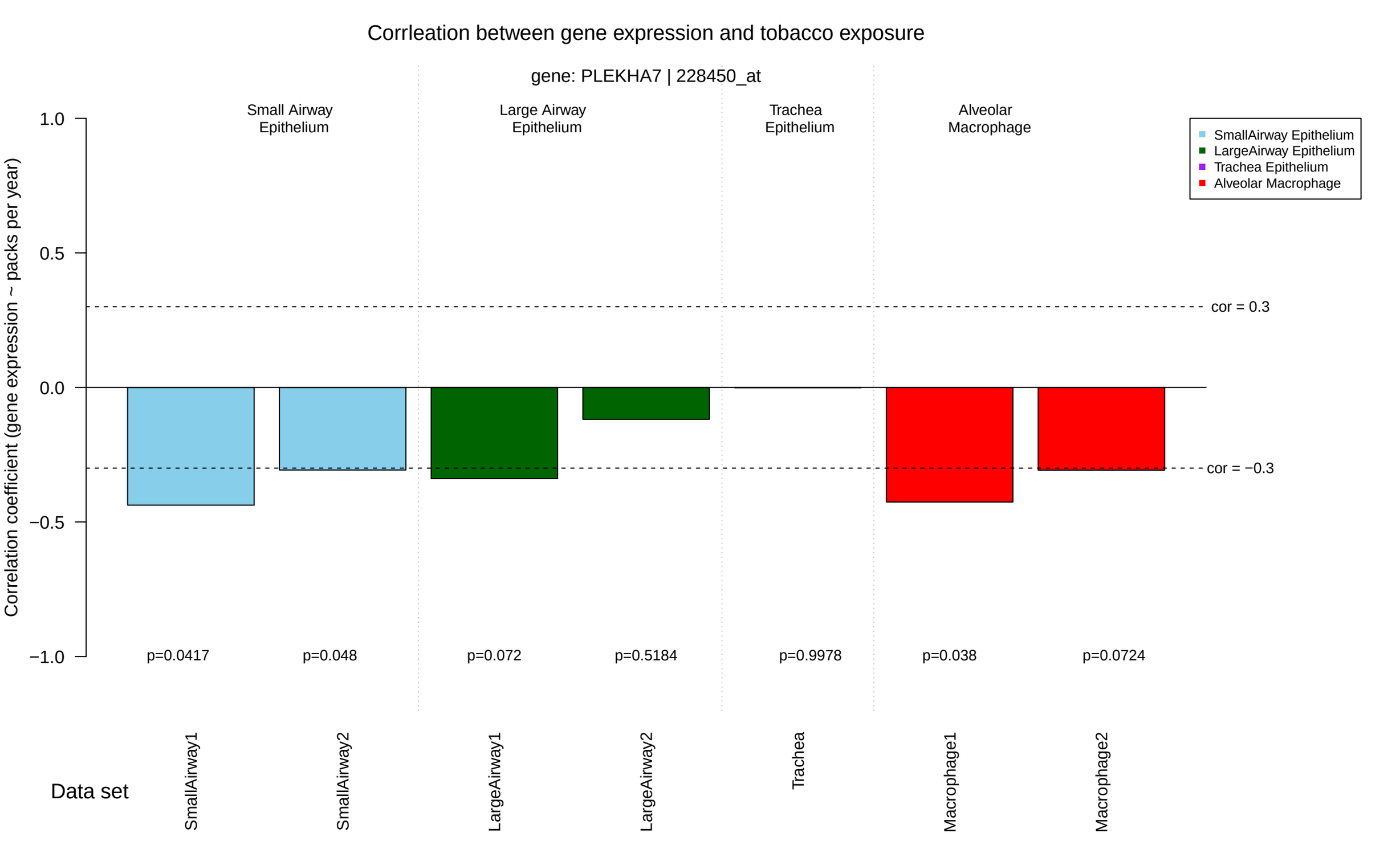

set: LargeAirway1 (p=0.01)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2424)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3422)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0119)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3319)
PLEKHA7 | 228450_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA7 | 228450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0417
p=0.048
p=0.072
p=0.5184
p=0.9978
p=0.038
p=0.0724
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2